

KF SURYA

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Work Experience

- **Gen AI Developer** Sep 2025 – Present
Connected Value Health Coimbatore, Tamil Nadu, India
 - Worked on **real-time healthcare project** building AI-powered solutions using FastAPI, LLMs, and Mistral OCR for automated medical document processing and information extraction.
 - Developed **production-grade RAG systems** with Vector Databases (Milvus, Pinecone) for intelligent healthcare data retrieval and personalized patient information delivery.
- **AI Intern** Jul 2024 – Aug 2024
Ei Systems Ghaziabad, Delhi NCR, India
 - Designed the **MediSaga RAG Chatbot** using RAPTOR indexing with Milvus and Llama2 for medical QA.
 - Engineered a **CPU-optimized RAG system** using quantized Llama2, improving inference efficiency without GPU.
- **ML Intern** Dec 2023 – Jan 2024
Yanne Technologies Coimbatore, Tamil Nadu, India
 - Developed a **LangChain-based Medical Assistant** using GPT-3.5 for accurate medical information retrieval.
 - Built a **Checkbox Detection app** using OpenCV for Vizhi Eye Care NGO, improving automation in form analysis.

Education

- **KPR Institute of Engineering and Technology** 2022 – Present
B.Tech in Artificial Intelligence and Data Science Coimbatore, India
 - CGPA: 8.0/10

Technical Projects

- **Multimodal RAG Chatbot** GitHub
FastAPI, Streamlit, ChromaDB, Vosk, Gemini 2.5 Flash, Docker
 - Built a **multimodal RAG system** supporting PDF, audio, and video inputs for context-aware AI responses.
 - Integrated **Gemini 2.5 Flash** and **Vosk speech-to-text** with **Dockerized deployment** for scalability.
- **Text-to-SQL Assistant** GitHub
Python, Chainlit, Gemini 2.5 Flash, SQLite
 - Developed an **AI-powered Text-to-SQL tool** converting natural language to SQL with Gemini 2.5 integration.
 - Implemented a **secure execution pipeline** with schema detection, validation, and real-time result display.
- **Company Expense Tracker** GitHub
Python 3.11, FastMCP, Gemini 2.5 Flash, SQLite
 - Built a **Model Context Protocol (MCP) server** enabling AI-driven expense and employee management.
 - Added **Gemini-based analytics** and safe CRUD operations with duplicate detection and cascade delete.
- **MediSaga — RAG Chatbot Application** GitHub
RAPTOR Indexing, Milvus, Llama2, Streamlit
 - Created a **medical QA chatbot** using RAPTOR indexing with Milvus and Llama2 for accurate retrieval.
- **Automatos — Automated Surveillance System** GitHub
Python, YOLOv5/8, OpenCV
 - Built an **automated surveillance pipeline** detecting individuals in CCTV feeds using YOLO models.

Technical Skills

- **Programming Languages:** Python, JavaScript, Java, C, SQL, Bash
- **AI & Machine Learning:** Retrieval-Augmented Generation (RAG), LLM Integration, Transformers, Model Quantization, Fine-Tuning
- **NLP & Multimodal AI:** LangChain, Llama2/3, Gemini, Sentence-Transformers, ChromaDB, FAISS, Vosk
- **Frameworks & Tools:** FastAPI, Streamlit, Chainlit, TensorFlow, PyTorch, OpenAI API, Google Gemini API, Docker, Conda, UV
- **Databases:** MongoDB, PostgreSQL, SQLite, Milvus, ChromaDB
- **Data Engineering & Cloud:** Pandas, NumPy, Matplotlib
- **Computer Vision:** OpenCV, YOLOv5/v8, Image Segmentation, Object Tracking
- **DevOps & Version Control:** Git, GitHub, Docker Compose, Environment Automation

Certifications & Awards

- **Best Paper Award (AMSAS-2022)** — Certified by DRDO & Dept. of Chemistry, KPRIET
- Machine Learning with Python — IBM (Honors)
- Exploratory Data Analysis — Coursera
- Python for Data Analysis — Coursera
- Git and GitHub — 365 Data Science